

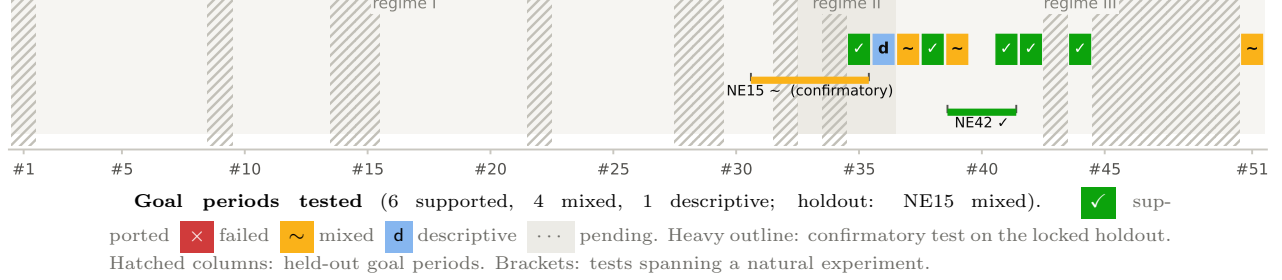
H05: Cutting the cross-room channel lowers entropy production, and rooms become coupled blocks

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 55% Faithfulness 2/5 Usefulness 2.5/5

Bottom line. Rooms cut chat coupling only through what agents read: on corrected data the room effect is stronger, survives day-edge trimming, and vanishes once ledger read counts are known; work is not room-structured, and the holdout verdict (on the buggy table) stays inconclusive.

level: hypothesis · pipeline checks 100% · updated 2026-10-04



WHAT HAPPENED

Do rooms make agents coupled blocks? Round 1 (exploratory, 99 non-holdout days, #35–#44) fitted 1-min talk and activity spins. Within-room talk coupling beat cross-room in 6/6 regime-III two-room windows (kinetic-Ising J , Fisher $p < 10^{-4}$), but per period the primary κ_x test passed only in G38, G41, G44 (mixed G37, G42; failed G39). The pooled two-way-FE effect of co-location was $\beta \approx +0.008$ ($z = 2.6$). Activity spins: not robust. Entropy production stayed at the cross-day null in 15/18 windows, so it cannot be measured at this resolution. Holdout, NE15 (03-16 #best/#rest split): the primary pair DiD passed (talk $\kappa_x -0.022 [-0.038, -0.008]$, $p_{\text{perm}} = 0.02$) and J_{out} fell to ≈ 0 . The pooled TWFE missed ($z = 1.4$), so the pre-registered verdict is **inconclusive**. Unpredicted: within-room J_{in} rose about $\times 6$. Caveat: the post-split window (#35) was not held out and had already been seen in round 1.

PHYSICAL MODEL

Kinetic Ising (model 02). $s_i(t) = \pm 1$: agent i talked (talk spin) or acted (activity spin) in 1-min bin t ; $s'_i = s_i(t+1)$. h_i : daily-schedule field (removed by a cross-day surrogate). r_i : room; J_{in} , J_{out} : same- and cross-room coupling; a cut predicts $J_{\text{out}} \rightarrow 0$. Σ_g : Aguilera–Ito–Kolchinsky lower bound on the entropy production σ , dual weights θ . $\hat{R}$, D : block correlations, variances. α_{ij} , γ_d : pair, day effects.

$$P(s'_i | s) = \frac{e^{s'_i H_i}}{2 \cosh H_i}, \quad H_i = h_i + \sum_j J_{ij} s_j$$

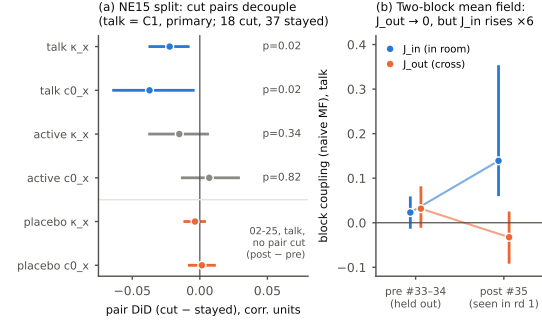
$$J_{ij} = J_{\text{in}} \delta_{r_i r_j} + J_{\text{out}} (1 - \delta_{r_i r_j})$$

$$\sigma = \sum_{i < j} (J_{ij} - J_{ji}) \langle g_{ij} \rangle, \quad g_{ij} = s'_i s_j - s_i s'_j$$

$$\Sigma_g = \max_{\theta} [\theta \cdot \langle g \rangle - \ln \langle e^{-\theta \cdot g} \rangle] \leq \sigma$$

$$J^{\text{MF}} = D^{-1/2} (I - \hat{R}^{-1}) D^{-1/2}$$

$$\kappa_{ij,d} = \alpha_{ij} + \gamma_d + \beta \text{coloc}_{ij,d}$$



Observables. (a) Confirmatory pair difference-in-differences at the 03-16 split: the coupling change of the 18 pairs put in different rooms minus that of the 37 pairs that stayed together (pre #33–34, held out; post #35, not held out). Talk coupling dropped (C1 passed); activity did not. The 02-25 placebo, which separated no one, shows no change. Bars: day-bootstrap 95% CIs; p : assignment permutation. (b) Block mean field: cross-room J_{out} falls to ≈ 0 , while within-room J_{in} rises (unpredicted).

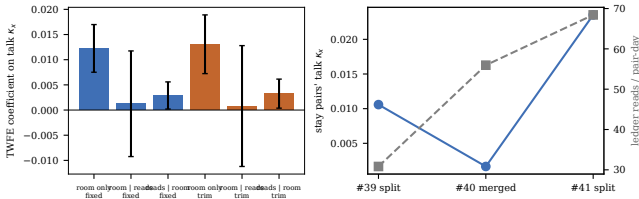
SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

Candidly, agents who can't see each other's chat stop timing their chat together: close to a restatement of the scaffold, since rooms are a visibility filter. Three things are mildly useful. (1) A cheap monitor: a pair DiD on 1-min talk spins caught the cut with 18 cut pairs and 5 post-split days (power only about 25–60%). (2) A negative result an operator splitting teams should know: activity, the actual work, is *not* room-structured. Rooms reorganize conversation, not labor. (3) The unpredicted $\times 6$ rise in within-room coupling hints that attention is conserved and redirected to roommates (HH90, now tested by H18). The thermodynamic half, entropy production, is unmeasurable here and buys nothing practical.

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RESULTS VS PREDICTIONS

Prediction	Round 1 → round 1b (fixed bins; trim)	Verdict (1b)
P2 talk within > cross	pooled +0.014 (5/6) → +0.021 (6/6, p 6e-7); trim same	yes
P2 per window	3/6 supported → 4/6 (G42 now; mostly G39 mixed)	
P2 active spins	4/6, p 0.01 → p 0.17; trim 2/6	no (with-drawn)
P3 pooled TWFE, talk	z 2.6 → 4.1; trim 3.3	yes
P3 split 05-11 DiD < 0	-0.008 n.s. → -0.020 (p 0.03)	yes
MF1 talk $J_{in} > J_{out}$, $J_{out} \ni 0$	6/6, 6/6 → 6/6, 4/6; trim 5/6, partly 3/6	
HH248: reads absorb the room	$\beta(\text{room})$ +0.012 (z 5.1) → +0.001 (z 0.2) given reads	yes (new)
N2 #focus cut	ledger reads -95%; talk DiD as predicted -0.006 n.s.	
N3 NE42 attention reallocation	stay pairs: merged -0.010, split +0.022 [0.013, 0.032]	yes (new)
Holdout C1 / C3 / C5	pass / fail / mixed (buggy table)	not re-run



Second figure. (a) Two-way FE coefficients on talk coupling, regime III non-holdout pair-days: co-location alone, co-location given $\log(1+\text{ledger reads})$, and reads given co-location (blue fixed bins, orange day-edge trimmed; bars 95%). (b) NE42: talk coupling (left) and ledger reads (right) of the 42 pairs that stayed together through the merge week.

GOAL PERIODS IN DETAIL

NE15 ~ 1b: mixed (not re-run; corrected non-holdout analogue... (**holdout**)

G35 ✓ 1b: supported p 0.004; fixed bins)

G36 d 1b: descriptive p 0.12; regime II)

G37 ~ 1b: mixed p 0.11)

G38 ✓ 1b: supported p 0.007; trim +0.018)

G39 ~ 1b: mixed +0.000; was failed)

G41 ✓ 1b: supported p 0.001)

G42 ✓ 1b: supported p 0.017; was mixed)

G44 ✓ 1b: supported p 0.001)

G51 ~ 1b: mixed (the ledger confirms the cut; talk DiD righ...

+1 more periods: see the card.

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	1
B assumptions	1	G ground truth	1
C adequacy	1	H comparative	0
D unfitted predictions	0	I transfer	1
E interventional	1		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writup/paper.tex`).

CAVEATS

Co-location and log reads correlate 0.82, so HH248's mediation has little leverage ($SE \times 2.2$). NE42's merge week carries a shared objective. The holdout run used the buggy table; the corrected non-holdout analogues suggest attenuation, not a sign error, but only a re-run can say. EP remains unmeasurable.

ROUND 2

Ask Vivian whether to re-run the NE12/NE15 holdout on the fixed panel (same predictions). Test HH248 on reply probability per read (DQ2).

- **Essence.** Through which channels does information actually move?
- **H05-R1** Rooms matter only insofar as they also split artifacts: cross-room pairs sharing a repo stay coupled through edits (E3).
- **H05-R2** The $\times 6$ rise in within-room coupling after the split is attention reallocation, predicted in size by $N_{\text{room}}^{-0.6}$ (E5).
- **H05-R3** Information leaks across rooms through history search and artifacts at a measurable rate (a leak conductance).

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